

Topic-wise Notes from Transcript

1. Test Design Techniques

Definition: Methods used to design effective test cases and prepare test data.

Purpose:

- Reduce the amount of test data while ensuring maximum coverage.
- Improve efficiency and minimize redundancy in testing.

Key Benefits:

- Covers all functional areas with minimal data.
- Ensures edge cases and negative scenarios are tested.

Interview Importance: Highly important.

Possible Interview Questions:

- What are test design techniques in software testing?
- Why do we use test design techniques?
- Explain different types of test design techniques with examples.

2. Types of Test Design Techniques

(a) Equivalence Partitioning

Concept: Divide input data into partitions where test cases from each partition represent all values in the partition.

Example: Age input field: If valid range is 18-60, partitions could be:

- Below 18 (invalid)
- 18-60 (valid)
- Above 60 (invalid)

Interview Importance: Frequently asked.

Question Variations:

- What is equivalence partitioning?
- Give an example of equivalence partitioning in login functionality.

(b) Boundary Value Analysis

Concept: Test at the boundaries of input ranges because errors often occur there.

Example: For age 18-60: Test 17, 18, 60, 61.

Interview Importance: Very common.

Question Variations:

- Explain boundary value analysis with an example.
- Why is boundary value analysis important?

(c) Decision Table Testing

Concept: Represents combinations of conditions and actions in a tabular form.

Example: Online money transfer: Conditions like account approved, sufficient balance, OTP valid · Action: Transfer money.

Interview Importance: Often asked for complex scenarios.

Question Variations:

- What is decision table testing?
- How do you design test cases using decision tables?

(d) State Transition Testing

Concept: Tests changes in system states based on events or inputs.

Example: Login attempts: State 1: Logged out · Enter valid credentials · State 2: Logged in.